

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457292.8556 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2 +/- 0.018
Transit depth (Rp/Rs)^2	3.99 +/- 0.73 [%]
Orbital Inclination (inc)	86.47 +/- 0.72
Airmass coefficient 1 (a1)	1.009 +/- 0.021
Airmass coefficient 2 (a2)	-0.006 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.12 %
Best Comparison Star	#2 - [216, 233]
Optimal Aperture	2.48
Optimal Annulus	11.07
Transit Duration (day)	0.0789 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2 ± 0.018 vs 0.1592 (deviation 2.27σ)
Duration fit vs published	0.0789 d vs 0.0946 d (deviation 1.62σ)
Mid-time O–C	-2.0 min
Depth SNR	11.1
Residual scatter	2.12 %

Light curve

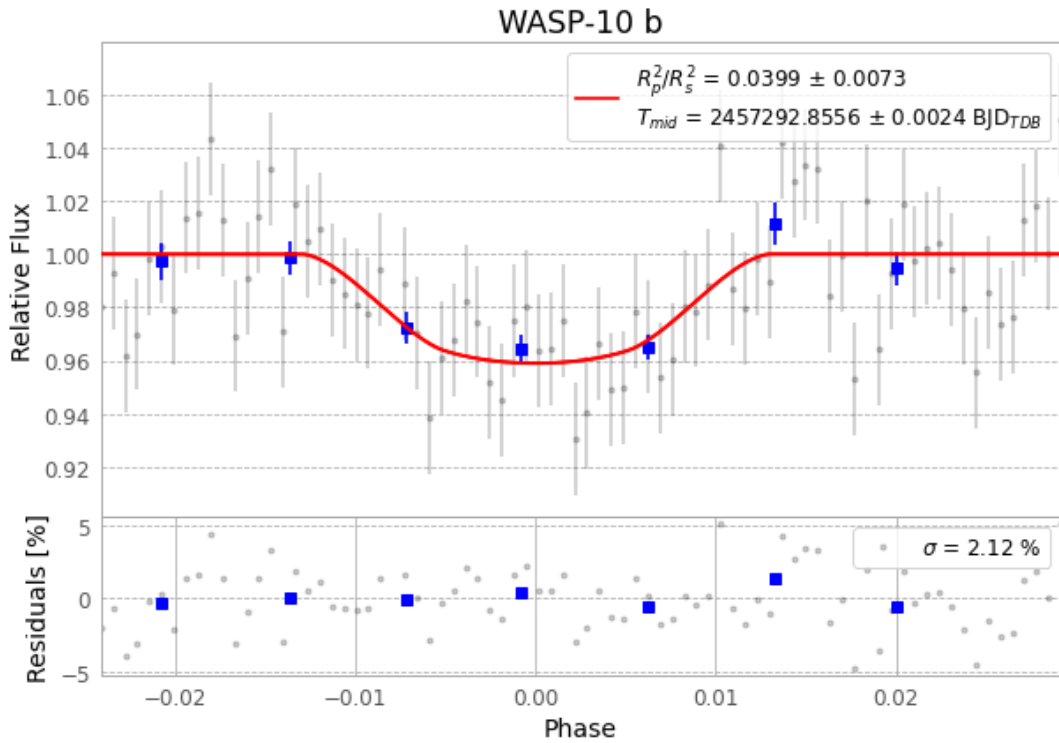


Figure 1 — FinalLightCurve_WASP-10 b_2015-09-26.png (SHA-256 d0f5d6b0eec570ec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee3698f2b596633b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), WASP-10150927063612.FITS → WASP-10150927103612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-150927.log (1f1cdc26f073...), FinalLightCurve_WASP-10 b_2015-09-26.png (d0f5d6b0eec5...), FinalLightCurve_WASP-10 b_2015-09-26.csv (bcb7216f8e6f...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:40Z.